

BusyBee: Contextual Inquiry Interviews

CSC 310 Human-Computer Interaction | Team: Omar Naguib, Nolan Jarvinen, Evan Wilkin

Each team member conducted a structured interview with real participants using the same ten questions below. Seven interviews were completed across the team. Responses are recorded as given.

Participants

Subject	Interviewed by
Subject 1	Omar Naguib
Subject 2	Omar Naguib
Subject 3	Evan Wilkin
Subject 4	Evan Wilkin
Subject 5	Evan Wilkin
Subject 6	Nolan Jarvinen
Subject 7	Nolan Jarvinen

Interview Questions

1. How often do you go out?
2. Do you plan ahead before going out, or are you more spontaneous?
3. How important is knowing an establishment's busyness to you?
4. What methods, if any, do you use to find the predicted or expected wait time?
5. What do you do when you can't tell how busy a place is going to be?
6. How often do you end up somewhere more crowded than you like?
7. How do you feel when a place is busier or has a longer wait than you expected?
8. How bad does the busyness or wait time need to be for you to consider leaving and trying another place?
9. Have you ever told someone else how busy a place was?
10. Would you trust a community-based predicted wait time or busyness app?

Subject 1

Question	Response
How often do you go out?	Most days, in some form. Being retired means the week doesn't have the shape it used to have. Groceries, the gym a couple of times, dinner out with my wife maybe twice. And there's the travel, which is the big one. Six or eight trips a year, seeing family or just going somewhere.
Do you plan ahead before going out, or are you more spontaneous?	Depends entirely on what it is. Dinner locally, we'll just go, and if there's a wait we'll go somewhere else or sit at the bar. No harm done. Flying, I plan it to the minute. I'm working backwards from boarding time the night before. There's nothing spontaneous about it, and I don't think I'd enjoy it if there was.
How important is knowing an establishment's busyness to you?	For a restaurant it's nice to know. It changes whether we go at six or at seven. It isn't going to ruin the evening either way. At an airport it's the single thing I'd most like to know and can't. Everything else about that morning I can control. The security line is the one part I'm guessing at, and it's the part that decides whether I make the flight.
What methods, if any, do you use to find the predicted or expected wait time?	For restaurants we'll just ring them and ask. Somebody picks up, tells you forty minutes, and that's that. For the airport there's the wait time they publish and I've stopped trusting it. It told me ten minutes once and I stood there the better part of forty. I don't know whether it was wrong or whether it was just old, and that's the part that bothers me. So mostly now I ask people. My wife goes through a different checkpoint and texts me which one is moving. The shuttle driver has done forty runs that morning, he knows which terminal is a mess.
What do you do when you can't tell how busy a place is going to be?	I add more time. That's the whole strategy, there isn't a second one. If it's a holiday, or I've got a connection I can't afford to miss, another forty five minutes on top. The cost of being early is a coffee and a magazine. The cost of being late is a missed flight, a rebooking fee and a night in a hotel. Those aren't the same size, so I don't treat them the same.
How often do you end up somewhere more crowded than you like?	More than I'd like to admit. A few times a year at the airport, where it actually matters. Locally, maybe once a month we'll walk somewhere on a Saturday and turn straight around. The airport ones stay with you though. The restaurant ones you've forgotten by the next day.
How do you feel when a place is busier or has a longer wait than you expected?	At a restaurant, mildly irritated, and then over it. At the airport it isn't irritation, it's a low level dread that builds the whole time you're standing there. You're doing the arithmetic over and over. If this line takes another fifteen minutes, where does that put me. And you can't do anything with the feeling, that's the worst part of it. You're just stuck in it.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	For dinner, twenty minutes and we'll look elsewhere. Thirty and we definitely will. At the airport there is no leaving. That's the whole point of it. You can't decide the line is too long and go to a different airport. You stand in it for however long it is. Which is exactly why I'd want to know before I left the house, because that's the only moment I've got any choice in it.
Have you ever told someone else how busy a place was?	All the time, to the family. We've got a group message and I'll send "leave now, security is a zoo" if I've just come through it. You've stood in it, you don't want the next person standing in it. Somewhere public, for strangers, I wouldn't think to. Who's reading it? What am I getting out of it?
Would you trust a community-based predicted wait time or busyness app?	More than the official number, I think, provided I could see when it was posted. Somebody who was standing there ten minutes ago is telling me something real. A sign is telling me what somebody typed in at some point. But I'd want to know how many people said it. One person could be at the wrong terminal, or just wrong. And if it was an hour old I'd ignore it, same as I ignore the signs now.

Subject 2

Question	Response
How often do you go out?	Every day basically, if going out counts as leaving the house. I'm on campus five days a week and I never study at home, so that's a cafe or the library most days. Then dinner with friends a couple of times a week, more at the weekend.
Do you plan ahead before going out, or are you more spontaneous?	For studying, zero planning. Class ends and I walk somewhere. The gap is however long it is, so thinking about it just eats into it. Going out with friends is the opposite, except it isn't really planning, it's negotiating. Six people in a group chat for twenty minutes. Someone suggests a place, someone checks Maps, someone says they went last Friday and it was awful.
How important is knowing an establishment's busyness to you?	Really important for studying, and I don't think I'd realised that until you asked. Busy means no table, and no table means the whole trip was pointless. For going out it matters in the opposite direction. If a place is dead on a Friday night I don't want to go. You want it to be busy, that's the whole point of going out.
What methods, if any, do you use to find the predicted or expected wait time?	Google Maps has the popular times thing, and it's a vibe. It's right about the general shape, lunch is busy obviously, but it doesn't know anything about today. It doesn't know it's midterms. During finals every cafe near the library is full at eleven at night and the graph says quiet, because normally it would be. Mostly I just message someone who's there. If someone from my course is at the library I'll ask if there's space, and that's better than any app because they're looking right at it. And sometimes I'll check whether a place got tagged on Instagram recently, you can tell from a story whether it's rammed.
What do you do when you can't tell how busy a place is going to be?	I just go. It's a five minute walk, and if it's full I'll go somewhere else. It's not a big deal any one time. Although if I actually think about it, that happens a lot, and it adds up. Twenty minutes here, twenty there. Some weeks I feel like I've spent half my study time looking for somewhere to study.
How often do you end up somewhere more crowded than you like?	Honestly, weekly. Probably more during midterms and finals. It's so normal that I don't really register it as a thing that's going wrong. It's just what happens.
How do you feel when a place is busier or has a longer wait than you expected?	Annoyed, but mostly at myself, which is stupid because there was no way for me to know. And if I'm on my own there's a bit of embarrassment in walking in, seeing it's full, and having to turn round and walk straight back out. Everyone watches you do it. It's not a big thing but I'd rather not.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	For studying it isn't about the wait at all, it's just binary. Either there's a table or there isn't. If there's nowhere to sit I've already left, there's nothing to wait for. For dinner with friends we'd wait half an hour easily, if it's somewhere we actually wanted to go. Longer if we've already walked there. And once six people have agreed on something, nobody wants to reopen it.
Have you ever told someone else how busy a place was?	Constantly. I'll take a photo of the place and send it to the group with something like "do not come here." That's just normal. I wouldn't write a review about it though. It's a thing you tell your friends, it isn't information, it's just complaining.
Would you trust a community-based predicted wait time or busyness app?	I'd try it. Whether I'd keep it depends on whether it's ever right when I check it, and I'd work that out within about two weeks. I'd trust it more than Maps because at least it's about today. But it only works if people actually use it. An app like that near campus with nobody posting is just an empty map, and I'd delete it. It would need to be quick, too. If reporting takes more than a couple of seconds I'm not doing it, I'm already walking.

Subject 3

Question	Response
How often do you go out?	Once every three weeks (sitting down, not ordering take out).
Do you plan ahead before going out, or are you more spontaneous?	Typically chooses what establishment to visit, but doesn't necessarily research where they want to go.
How important is knowing an establishment's busyness to you?	Medium. Nobody wants to wait or be crowded, but where they go, it isn't a huge concern.
What methods, if any, do you use to find the predicted or expected wait time?	Will look at the predicted busyness through Google, or if it is a movie theater will check how many seats are bought. Checks online casually and infrequently.
What do you do when you can't tell how busy a place is going to be?	Debates the desire versus the potential wait.
How often do you end up somewhere more crowded than you like?	Pretty rarely, about 4 times a year, due to the infrequency of going out (separate from daily highway traffic).
How do you feel when a place is busier or has a longer wait than you expected?	Moderately inconvenienced to discouraged.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	Any wait time is too much when going with a family with two young children.
Have you ever told someone else how busy a place was?	Yes, both when a friend is coming to meet up or when they are going separately, will let them know.
Would you trust a community-based predicted wait time or busyness app?	As a starting point, as long as there is active community use. Sees no reason for someone to rate a place not busy when it is. Already trusts apps such as GasBuddy.

Subject 4

Question	Response
How often do you go out?	3 to 4 times a day, most likely 3.
Do you plan ahead before going out, or are you more spontaneous?	Always plans ahead. Last thing they want is to walk around not knowing what they need.
How important is knowing an establishment's busyness to you?	Depends on the establishment. Doesn't want to go to the gym when it is packed since it's a smaller sized gym. Grocery stores are also a mess when busy, wants to get in and out.
What methods, if any, do you use to find the predicted or expected wait time?	Google predicted busyness.
What do you do when you can't tell how busy a place is going to be?	Just goes and risks it being busier than normal.
How often do you end up somewhere more crowded than you like?	Happens about 2 to 3 times a week. Even going to the gym at 6 a.m. is busier than expected.
How do you feel when a place is busier or has a longer wait than you expected?	Stressed about wasting time.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	10 minutes or more will cause them to leave.
Have you ever told someone else how busy a place was?	Maybe like once.
Would you trust a community-based predicted wait time or busyness app?	Doesn't know, maybe. Would have to trial and error it.

Subject 5

Question	Response
How often do you go out?	At least once or twice every week.
Do you plan ahead before going out, or are you more spontaneous?	Usually a plan ahead type of person.
How important is knowing an establishment's busyness to you?	Not super important. Nice to know ahead, but if it's busy when they get there, they'll just wait.
What methods, if any, do you use to find the predicted or expected wait time?	Usually just a Google based wait time.
What do you do when you can't tell how busy a place is going to be?	Still goes anyway. If it's too busy, just goes somewhere else.
How often do you end up somewhere more crowded than you like?	Pretty often, probably once every week.
How do you feel when a place is busier or has a longer wait than you expected?	It sucks but will put up with it, especially if it's a restaurant they want to eat at.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	15 minutes is the average wait time. If already seated, will wait 20 minutes for a waiter before trying to flag someone down, and will only leave if that doesn't work.
Have you ever told someone else how busy a place was?	Yes. If they are coming to meet up, they'll let them know before heading over. If someone asks about a restaurant, they'll give their prediction.
Would you trust a community-based predicted wait time or busyness app?	Yes, if the results are reliable and there is a community already using it.

Subject 6

Question	Response
How often do you go out?	I go out almost every day. Usually there is something I need to do, like going to a store or getting something to eat.
Do you plan ahead before going out, or are you more spontaneous?	I usually plan ahead before I go somewhere. I like knowing where I'm going and having an idea of how long I will be there.
How important is knowing an establishment's busyness to you?	Pretty important. I don't really like being around a lot of people, so if I know a place is going to be really crowded, I wouldn't typically go, or I would look for another place depending on where it is I would be going.
What methods, if any, do you use to find the predicted or expected wait time?	I don't really use anything right now to check wait times or how busy somewhere is before I go. Most of the time I just go there and find out once I arrive.
What do you do when you can't tell how busy a place is going to be?	I'll go to the business and see how busy it is when I get there.
How often do you end up somewhere more crowded than you like?	Probably more than half of the time that I go out. It happens pretty frequently, especially when I go somewhere at a time when a lot of other people are out too.
How do you feel when a place is busier or has a longer wait than you expected?	It makes me anxious, especially when a place is much more crowded than I was expecting.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	For a restaurant, if the wait is over about 30 minutes, I would probably consider going somewhere else. I also wouldn't really want to stay for a movie if the theater was extremely crowded. Stores are different though. If a store is busy, I'll typically still go in and get what I need.
Have you ever told someone else how busy a place was?	Probably. I've definitely talked about how busy places are. If I knew someone else was thinking about going there, I might tell them how crowded it was.
Would you trust a community-based predicted wait time or busyness app?	Probably, as long as the app was run well and the information seemed accurate. I already use Waze, which relies on others reporting things like traffic, accidents, and other conditions, so I could see myself using something similar to find out how busy a business is.

Subject 7

Question	Response
How often do you go out?	If we're counting anywhere or anything, I leave the house almost every day. I go to the gym six days a week, the store about once a week, and then I have work, or I'll drive all the way to Flint, so I'm usually going somewhere every day.
Do you plan ahead before going out, or are you more spontaneous?	I think it depends on what I have going on that day. If I'm doing something like golfing, I'll have a plan to go golfing, but whatever I do after might be spontaneous. For things like going to the grocery store or the gym, I usually know ahead of time that I'm going.
How important is knowing an establishment's busyness to you?	Somewhat important. If it's a restaurant, I like knowing whether or not it's busy, but for something like a grocery store, I don't care as much. If I'm going to downtown Detroit, it's more important because I want to know what traffic and parking are going to be like. It would also be nice to know if something like a golf course is really busy.
What methods, if any, do you use to find the predicted or expected wait time?	I'll usually check Google Maps or Apple Maps to see how long the drive is and what traffic looks like. For restaurants, I'll look online to see if there are reservations available or if they have a wait time posted. Sometimes I'll call ahead and ask how busy they are or if they have any tables open.
What do you do when you can't tell how busy a place is going to be?	Usually I'll just take my chances and go there. If I get there and it's too busy, I'll just go somewhere else.
How often do you end up somewhere more crowded than you like?	If we're talking about traffic, every day. For actual places, I'd say maybe 25% of the time. I'm usually pretty good at predicting when somewhere is going to be busy and trying to beat the crowd.
How do you feel when a place is busier or has a longer wait than you expected?	It's more of an "aw man" or "dang it" kind of feeling. Like I wanted to eat or hang out there, and now I might have to change plans. I don't know how to describe the emotion, really. It's annoying I guess.
How bad does the busyness or wait time need to be for you to consider leaving and trying another place?	For a restaurant, around 45 minutes is usually my cutoff, especially depending on how hungry I am. If I could get a drink or something while I waited, I might stay longer. For stores, it would have to be really busy. The Costco in Ann Arbor can get packed, and seeing a really full parking lot or really long checkout lines can get frustrating.
Have you ever told someone else how busy a place was?	Yeah. I can't really think of a specific time, but if I'm meeting someone somewhere and I get there first, I'll usually let them know if it's really busy. Then we can decide if we want to go somewhere else.
Would you trust a community-based predicted wait time or busyness app?	I would, but it would have to be done the right way. I'd want to know how they make sure the information is trustworthy. If the app explained how it deals with false reports, trolls, and moderation, I'd be more likely to trust what people are reporting.

Key Takeaways

1. **Trust in any busyness source is always conditional, never a blanket yes.** Every subject qualified their answer: a timestamp and a headcount of reporters (Subject 1), two weeks of proof plus enough nearby users (Subject 2), trial and error (Subject 4), a track record and an existing community (Subjects 3, 5), a well-run and accurate app (Subject 6), and an explanation of how false reports and trolls get handled (Subject 7).
2. **A person who is there right now is trusted over any app, sign, or published number.** Subject 1 asks his wife and a shuttle driver. Subject 2 messages someone already at the library. Subject 5 calls the restaurant directly. Subject 7 calls ahead to ask if tables are open.
3. **How much busyness matters depends on the stakes of that specific trip, not a fixed priority.** Subject 1 treats a restaurant wait as trivial and an airport line as the one thing that decides if he makes his flight. Subject 2 treats a full table as pass or fail for studying but wants a place busy when out with friends. Subject 7 cares for a restaurant or downtown parking, not for a grocery run. Subject 3 barely cares in general but raises the stakes explicitly around a family with young children.
4. **People already tell each other how busy a place is, but only people they already know, and they don't think of it as reporting.** Subject 1 texts family, not strangers (“who's reading it, what am I getting out of it”). Subject 2 posts photos to a group chat constantly but calls it complaining, not information. Subjects 3, 4, 5, and 7 all say yes, but only when meeting someone or asked directly.
5. **Existing tools are known to fail in specific ways, so people build their own workarounds instead of trusting them.** Subject 1's airport wait time said ten minutes and was forty. Subject 2's Google popular times “doesn't know it's midterms.” Subject 6 uses nothing right now and just finds out on arrival. Subject 7 pairs Maps traffic with a phone call because neither alone is enough.
6. **The cost of being wrong drives behavior more than how often it happens.** Subject 1 states it outright: a coffee and a wait versus a missed flight and a hotel room “aren't the same size.” Subject 2 treats a wasted trip to a full cafe as a binary loss. Subject 3 sets a lower tolerance specifically because of young kids in tow. Subject 6 will still enter a busy store to get what she needs but won't stay for a crowded movie.
7. **Reactions scale with stakes and visibility, from mild annoyance to real dread.** Subject 1 describes a low level dread and helplessness at the airport versus being mildly irritated at a restaurant. Subject 6 says it makes her anxious. Subject 2 is annoyed at herself and also embarrassed walking back out in front of others. Subject 4 feels stressed about wasting time. Subject 7's reaction is the mildest: “aw man, dang it.”

Affinity Diagram

Every response in the seven interviews above was broken into single-observation notes, written onto a note each, and grouped from the bottom up until the groupings stabilised. Group labels are phrased in the participants' own voice rather than as categories, following Beyer and Holtzblatt's contextual design method. The wall below holds **137 notes** in **42 groups** under **9 themes**, covering all ten questions and all seven participants.

Every note is reproduced here in full. The subject and question number on each note point back to the response it was coded from, which is printed verbatim on pages 2 to 8 of this document.

How to read a note

The security line is the only part of the morning I'm guessing at.

● S1 · Q3

The note

The colour

S1 · Q3

recorded

one observation, in the participant's own words, never two

which of the seven subjects it came from

Subject 1, question 3, so the full response can be checked

marks the three interviews captured as the interviewer's summary rather than verbatim

The seven subjects, and their colour on the wall

Subject	Who they are	Interviewed by	Capture
 Subject 1	Retired, six to eight flights a year	Omar Naguib	verbatim
 Subject 2	Student, studies off-campus every day	Omar Naguib	verbatim
 Subject 3	Dines out every three weeks, two young children	Evan Wilkin	recorded
 Subject 4	Out three to four times a day, errand-driven	Evan Wilkin	recorded
 Subject 5	Out once or twice a week, patient planner	Evan Wilkin	recorded
 Subject 6	Out most days, strongly crowd-averse	Nolan Jarvinen	verbatim
 Subject 7	Out daily: gym, stores, golf, long drives	Nolan Jarvinen	verbatim

The nine themes

- T1** How much busyness matters is set by the trip, not by the person
- T2** The tools I have describe last Tuesday, not today
- T3** A person who is standing there beats any app
- T4** With no information, I absorb the risk myself
- T5** Being wrong costs composure, not just minutes
- T6** Everyone has a walk-away number, and everyone's is different
- T7** The only useful moment is before I leave, and I'm already moving
- T8** People already warn each other, they just don't call it reporting
- T9** Trust is conditional, and not one participant said yes outright

T1 How much busyness matters is set by the trip, not by the person

WHAT THIS MEANS FOR BUSYBEE

Busyness is not one number. A full cafe is a **failed trip** for a student and a **good sign** for a Friday night out. Let a report carry the venue type and let the user state which direction they want. BusyBee's vibe tags are the hook for this, and the gauge should read against the user's intent, not an absolute scale.

The same person cares completely differently depending on the venue

A restaurant wait only changes whether we eat at six or at seven.

● S1 · Q3

At an airport it is the one thing I most want to know and cannot get.

● S1 · Q3

The security line is the only part of the morning I'm guessing at.

● S1 · Q3

Restaurant yes, grocery store no.

● S7 · Q3

Downtown matters because of traffic and parking, not the venue itself.

● S7 · Q3

It would be nice to know if a golf course is really busy.

● S7 · Q3

Won't go to the gym when it's packed, because it's a small gym.

● S4 · Q3 · recorded

Grocery stores are a mess when busy; wants in and out.

● S4 · Q3 · recorded

A busy store still gets my business; a busy theater doesn't.

● S6 · Q8

Wouldn't stay for a movie if the theater were extremely crowded.

● S6 · Q8

For a movie theater I check how many seats have been bought.

● S3 · Q4 · recorded

Who I'm with resets my limit before I even arrive

Any wait at all is too much with two young children along.

● S3 · Q8 · recorded

Once six people have agreed on somewhere, nobody reopens it.

● S2 · Q8

We'll wait longer if we've already walked there.

● S2 · Q8

My cutoff moves with how hungry I am.

● S7 · Q8

If I could get a drink while I waited, I'd stay longer.

● S7 · Q8

I'll put up with it if it's a restaurant I actually want.

● S5 · Q7 · recorded

Sometimes I want the place to be busy

A place that's dead on a Friday night is the wrong place.

● S2 · Q3

Busy means no table, and no table means the whole trip was pointless.

● S2 · Q3

I hadn't realised how much studying depends on it until asked.

● S2 · Q3

T1 How much busyness matters is set by the trip, not by the person

continued

And the baseline weight varies a lot from person to person

Pretty important, because I don't like being around a lot of people.

● S6 · Q3

Medium. Nobody wants to wait, but it isn't a huge concern.

● S3 · Q3 · recorded

Not super important. Nice to know, but I'll just wait.

● S5 · Q3 · recorded

Somewhat important, and entirely situational.

● S7 · Q3

T2 The tools I have describe last Tuesday, not today

WHAT THIS MEANS FOR BUSYBEE

Every score needs a **visible age**. The competitor is a historical average, so BusyBee wins by never showing one: live-only reports with a 30-minute decay, and a timestamp shown as prominently as the score itself.

Google's graph knows the pattern, not the day

Popular times is a vibe: right about the shape, blind to today.

● S2 · Q4

It doesn't know it's midterms.

● S2 · Q4

Google predicted busyness is the only thing I use.

● S4 · Q4 · recorded

Usually just a Google-based wait time.

● S5 · Q4 · recorded

I check Google's predicted busyness casually and infrequently.

● S3 · Q4 · recorded

The official number burned me once and I stopped believing it

The published airport wait said ten minutes. I stood there forty.

● S1 · Q4

I never found out whether it was wrong or just old, and that's what bothers me.

● S1 · Q4

An hour-old number gets ignored, same as I ignore the signs now.

● S1 · Q10

So I improvise, or I use nothing at all

I don't use anything. I go and find out when I arrive.

● S6 · Q4

I check Instagram stories, because you can tell if a place is rammed.

● S2 · Q4

Maps for traffic, the website for a posted wait, then a phone call.

● S7 · Q4

No single source is enough on its own.

● S7 · Q4

T3 A person who is standing there beats any app

WHAT THIS MEANS FOR BUSYBEE

Frame a report as **eyewitness testimony, not prediction**. Lead with the human fact, such as “4 people reported from here in the last 12 minutes”, and keep the aggregation logic out of the way. The product BusyBee is replacing is a text message from a friend.

I message whoever is already inside

Someone at the library beats any app, because they're looking right at it.

● S2 · Q4

My wife texts me which checkpoint is actually moving.

● S1 · Q4

I ring the place and ask a human

Somebody picks up, tells you forty minutes, and that's that.

● S1 · Q4

I call ahead and ask if they have tables open.

● S7 · Q4

I trust people whose job keeps them there all day

The shuttle driver has done forty runs, so he knows which terminal is a mess.

● S1 · Q4

What I'm buying is recency, not authority

Someone standing there ten minutes ago is telling me something real.

● S1 · Q10

I'd trust it more than Maps because at least it's about today.

● S2 · Q10

If someone asks about a restaurant, I'll give them my prediction.

● S5 · Q9 · recorded

T4 With no information, I absorb the risk myself

WHAT THIS MEANS FOR BUSYBEE

The behavior BusyBee replaces is **padding and gambling**, not searching. Success is a trip not wasted, so the confirmation screen should tell the reporter what their report prevented, and the app should be measured in avoided trips rather than sessions.

I pad the schedule and eat the cost

I add more time. That's the whole strategy, there isn't a second one.

● S1 · Q5

Holiday or a tight connection means another forty-five minutes on top.

● S1 · Q5

Early costs a coffee. Late costs a flight, a rebooking and a hotel.

● S1 · Q5

T4 With no information, I absorb the risk myself

continued

Or I just go and find out

I go to the business and see how busy it is when I get there.

● S6 · Q5

I take my chances; if it's too busy I go somewhere else.

● S7 · Q5

Just goes and risks it being busier than normal.

● S4 · Q5 · recorded

Still goes anyway; if it's too busy, goes somewhere else.

● S5 · Q5 · recorded

It's a five-minute walk. Not a big deal any one time.

● S2 · Q5

I weigh how much I want it against the wait it might have.

● S3 · Q5 · recorded

I time-shift to beat the crowd, and it still fails

I'm pretty good at predicting busy times and beating the crowd.

● S7 · Q6

Even the gym at 6 a.m. is busier than expected.

● S4 · Q6 · recorded

It happens when I go at a time a lot of other people are out.

● S6 · Q6

How often the guess goes wrong (self-reported)

More than half the time I go out.

● S6 · Q6

Two or three times a week.

● S4 · Q6 · recorded

Pretty often, about once every week.

● S5 · Q6 · recorded

Weekly, and more during midterms and finals.

● S2 · Q6

About 25% of the time for places; for traffic, every day.

● S7 · Q6

A few times a year at the airport; about monthly locally.

● S1 · Q6

Rarely, about four times a year, because I rarely go out.

● S3 · Q6 · recorded

Each failure is trivial, so the total stays invisible

Twenty minutes here, twenty there, and some weeks it's half my study time.

● S2 · Q5

It's so normal I don't register it as something going wrong.

● S2 · Q6

The airport ones stay with you. The restaurant ones are gone by morning.

● S1 · Q6

T5 Being wrong costs composure, not just minutes

WHAT THIS MEANS FOR BUSYBEE

Tone must scale with stakes. A restaurant deserves the calm half-circle gauge; a high-consequence context deserves unambiguous lead-time language and a bias toward over-warning. And because people **blame themselves** for a wasted trip, the empty state should say the data isn't there yet, and never imply the user missed something.

Low stakes produce annoyance that passes

It's an "aw man" or "dang it" kind of feeling.

● S7 · Q7

I can't describe the emotion. It's annoying, I guess.

● S7 · Q7

At a restaurant, mildly irritated, and then over it.

● S1 · Q7

Moderately inconvenienced to discouraged.

● S3 · Q7 · recorded

It sucks, but I put up with it.

● S5 · Q7 · recorded

High stakes produce dread, anxiety and stress

Not irritation, but a low-level dread that builds while you stand there.

● S1 · Q7

You do the arithmetic over and over: another fifteen minutes, then what?

● S1 · Q7

It makes me anxious when it's much more crowded than I expected.

● S6 · Q7

Stressed about wasting time.

● S4 · Q7 · recorded

The helplessness is the worst part of it

You can't do anything with the feeling. You're just stuck in it.

● S1 · Q7

At the airport there is no leaving. That's the whole point of it.

● S1 · Q8

There's a social sting on top of the lost time

Walking in, seeing it's full, and walking straight back out while everyone watches.

● S2 · Q7

Annoyed at myself, which is stupid, because there was no way to know.

● S2 · Q7

T6 Everyone has a walk-away number, and everyone's is different

WHAT THIS MEANS FOR BUSYBEE

The walk-away number spans **0 to 45 minutes** across seven people. A single “Busy” badge can't serve that range, so let users set a personal threshold and alert against it, and support a **capacity state** (“no tables”) alongside a wait in minutes, because for study trips the wait is irrelevant.

My cutoff, stated in minutes

Ten minutes or more and I leave.

● S4 · Q8 · recorded

Fifteen minutes is about my average wait.

● S5 · Q8 · recorded

Twenty and we look elsewhere; thirty and we definitely will.

● S1 · Q8

Half an hour easily, for somewhere we actually wanted to go.

● S2 · Q8

Over about thirty minutes and I'd consider somewhere else.

● S6 · Q8

Around forty-five minutes is usually my cutoff.

● S7 · Q8

Any wait at all, with two young kids in tow.

● S3 · Q8 · recorded

For some trips it isn't a wait at all, it's pass or fail

Either there's a table or there isn't. There's nothing to wait for.

● S2 · Q8

And sometimes there is no walking away

You stand in it for however long it is.

● S1 · Q8

If a store is busy I still go in and get what I need.

● S6 · Q8

The clock keeps running after I'm seated

Twenty minutes for a waiter, then I flag someone down.

● S5 · Q8 · recorded

I read the crowd from outside before I commit

A full parking lot and long checkout lines at Costco are the signal.

● S7 · Q8

T7 The only useful moment is before I leave, and I'm already moving

WHAT THIS MEANS FOR BUSYBEE

Two moments carry the product: **pre-departure** and **on arrival**. Reading must work before leaving the house; reporting must be a single tap from a notification or widget in under two seconds, with the slider, tags and note all strictly optional.

How often the question even comes up

Three to four times a day, most likely three.

● S4 · Q1 · recorded

Almost every day: gym six days a week, store weekly, work, long drives.

● S7 · Q1

Almost every day, usually a store or getting something to eat.

● S6 · Q1

Every day. Campus five days a week, and I never study at home.

● S2 · Q1

Most days, plus six or eight trips a year.

● S1 · Q1

At least once or twice every week.

● S5 · Q1 · recorded

Once every three weeks, sitting down.

● S3 · Q1 · recorded

The decision window closes when I set off

I'd want to know before I left the house, because that's my only moment of choice.

● S1 · Q8

Always plans ahead, and hates walking around not knowing what they need.

● S4 · Q2 · recorded

I like knowing where I'm going and how long I'll be there.

● S6 · Q2

Usually a plan-ahead type of person.

● S5 · Q2 · recorded

But plenty of trips contain no planning at all

For studying, zero planning. Class ends and I walk somewhere.

● S2 · Q2

The gap is however long it is, so thinking about it eats into it.

● S2 · Q2

Dinner locally we just go; flying I plan to the minute.

● S1 · Q2

Golf is planned; whatever I do after might be spontaneous.

● S7 · Q2

Chooses the establishment, but doesn't research it.

● S3 · Q2 · recorded

Choosing where to go is a group negotiation, not a plan

Six people in a group chat for twenty minutes, not planning but negotiating.

● S2 · Q2

Someone suggests a place, someone checks Maps, someone says it was awful.

● S2 · Q2

T7 The only useful moment is before I leave, and I'm already moving

continued

If contributing costs me anything, I skip it

More than a couple of seconds and I'm not doing it, because I'm already walking.

● S2 · Q10

Wants to get in and get out.

● S4 · Q3 · recorded

The second useful moment is on arrival, for the people behind me

I get there first, tell them it's busy, and we decide together.

● S7 · Q9

T8 People already warn each other, they just don't call it reporting

WHAT THIS MEANS FOR BUSYBEE

The reporting behavior exists, it's just trapped in group chats. Two moves unlock it: give a report a **share-to-chat path** so posting publicly costs nothing extra, and close the loop with a **reciprocity receipt** ("you kept 12 people out of that line") rather than abstract points, because "what am I getting out of it" is the real objection.

I tell my group the moment I see it

"Leave now, security is a zoo" goes to the family group message.

● S1 · Q9

I photograph the place and send "do not come here." That's just normal.

● S2 · Q9

But only for people I'm meeting, or people who ask

Yes, when a friend is meeting up or going separately.

● S3 · Q9 · recorded

If they're coming to meet up I'll tell them before they head over.

● S5 · Q9 · recorded

If I knew someone was thinking of going, I might tell them.

● S6 · Q9

Yeah. Can't name a time, but if I'm there first I say something.

● S7 · Q9

I don't think of what I'm sending as information

I wouldn't write a review. It isn't information, it's complaining.

● S2 · Q9

T8 People already warn each other, they just don't call it reporting

continued

Warning strangers has no obvious payoff

Somewhere public, for strangers, I wouldn't think to. Who's reading it?

● S1 · Q9

Maybe like once.

● S4 · Q9 · recorded

The motive is already there, and it's reciprocity

You've stood in it. You don't want the next person standing in it.

● S1 · Q9

T9 Trust is conditional, and not one participant said yes outright

WHAT THIS MEANS FOR BUSYBEE

Trust signals are **features, not polish**. Ship the timestamp, the reporter count and the decay on the face of the score; write a plain-language page on how false reports are handled and link it from the gauge; and seed venues so no one's first open is an empty map. Name Waze and GasBuddy in onboarding, since two participants reached for them unprompted.

Show me when it was posted

More than the official number, provided I can see when it was posted.

● S1 · Q10

An hour old and I ignore it, same as I ignore the signs now.

● S1 · Q10

Show me how many people said it

One person could be at the wrong terminal, or just wrong.

● S1 · Q10

Tell me how you handle liars and trolls

I'd want to know how they make sure the information is trustworthy.

● S7 · Q10

Explain false reports, trolls and moderation and I'd trust it more.

● S7 · Q10

I see no reason for someone to rate a place not busy when it is.

● S3 · Q10 · recorded

It has to be right the times I actually check

I'd know within about two weeks whether to keep it.

● S2 · Q10

Probably, as long as it's run well and the information seems accurate.

● S6 · Q10

Yes, if the results are reliable.

● S5 · Q10 · recorded

T9 Trust is conditional, and not one participant said yes outright

continued

Doesn't know, maybe. Would have to trial and error it.

● S4 · Q10 · recorded

An empty map is worse than no app at all

Nobody posting near campus and it's just an empty map. I'd delete it.

● S2 · Q10

As a starting point, as long as there's active community use.

● S3 · Q10 · recorded

Yes, if there's a community already using it.

● S5 · Q10 · recorded

I already trust apps built exactly this way

I use Waze, which relies on other people reporting conditions.

● S6 · Q10

Already trusts apps such as GasBuddy.

● S3 · Q10 · recorded

Personas

Three personas were built from the affinity wall, not from imagination. Each one is assembled from named subjects and named themes, and every goal, frustration and behaviour below traces back to a note on the wall. Between them the three account for all seven participants, with no participant used twice.

The names Maya, Jordan and Sam carry over from the team's earlier research pass so that the deliverables stay consistent with one another. The substance was re-derived from these interviews, and where the interviews contradicted the earlier draft the interviews won. Sam is the clearest case, and the change is recorded on his page.

The three at a glance

Persona	Built from	Themes	What decides it for them
Maya The high-stakes planner	Subjects 1 and 4	T1, T4, T5, T6, T7	The cost of being wrong is asymmetric, and she budgets against the expensive side of it.
Jordan The spontaneous local	Subjects 2 and 7	T1, T2, T4, T5, T7	Each wasted trip is too small to complain about, which is exactly why the total goes unnoticed.
Sam The reciprocal reporter	Subjects 3, 5 and 6	T3, T8, T9	He contributes because he has stood in the line himself, not because anyone is keeping score.

Each persona page below carries the same fields: a snapshot of how they move through a week, what they are trying to achieve, what currently goes wrong, what they already do about it, the point at which they walk away, what would earn their trust, and what BusyBee has to do to serve them.



Maya The high-stakes planner

Built from Subjects 1 and 4 · Themes T1, T4, T5, T6, T7

SNAPSHOT

Flies six to eight times a year and plans those mornings to the minute, working backwards from boarding time the night before. Everything else she does is casual. The difference isn't the venue, it's whether a deadline is attached.

GOALS

- Know before leaving the house, because that is the last moment a choice exists
- Pick the checkpoint or lane that is actually moving, not the nearest one
- Stop paying a 45-minute padding tax on every trip that matters

FRUSTRATIONS

- Published wait times are stale, and there is no way to tell stale from wrong
- Once she is in the line there is no leaving, because the decision was already made
- The waiting itself is dread plus arithmetic, not boredom

WHAT THEY ALREADY DO

- Adds more time. "That's the whole strategy, there isn't a second one."
- Asks people who are physically there: her wife at another checkpoint, the shuttle driver
- Texts the family group the moment she clears a line

"The cost of being early is a coffee and a magazine. The cost of being late is a missed flight, a rebooking fee and a night in a hotel. Those aren't the same size."
Subject 1

WHERE THEY WALK AWAY

20 to 30 min for dinner, but where it matters, no threshold at all

WHAT EARNS THEIR TRUST

Needs a visible timestamp and a count of how many people reported. Ignores anything an hour old.

WHAT BUSYBEE HAS TO DO

FlightSync puts a live checkpoint and lane breakdown inside the flight lookup she is already doing, and shows the score's age next to the score itself.

NOTES ON THE WALL BEHIND THIS PERSONA

A restaurant wait only changes whether we eat at six or at seven.

● S1 · Q3

I add more time. That's the whole strategy, there isn't a second one.

● S1 · Q5

At a restaurant, mildly irritated, and then over it.

● S1 · Q7

Ten minutes or more and I leave.

● S4 · Q8 · recorded

Three to four times a day, most likely three.

● S4 · Q1 · recorded

At an airport it is the one thing I most want to know and cannot get.

● S1 · Q3



Jordan The spontaneous local

Built from Subjects 2 and 7 · Themes T1, T2, T4, T5, T7

SNAPSHOT

Out most days with no plan attached. Studies off campus every day, eats out with friends several times a week. Decisions get made while already walking, or negotiated by six people in a group chat.

GOALS

- Know whether there is a seat before committing to the walk
- Avoid walking in, seeing it full, and walking straight back out
- Settle the group chat faster than twenty minutes

FRUSTRATIONS

- Popular times knows the pattern but “doesn't know it's midterms”
- Blames herself for a wasted trip she had no way to predict
- Turning around in a full room is a small public humiliation

WHAT THEY ALREADY DO

- Just goes, because it's a five minute walk and it's not a big deal any one time
- Messages whoever is already inside, because they are looking right at it
- Reads Instagram stories to see whether a place is rammed

“Some weeks I feel like I've spent half my study time looking for somewhere to study.”

Subject 2

WHERE THEY WALK AWAY

Binary when studying: a table or no table. Half an hour socially.

WHAT EARNS THEIR TRUST

Will try it and decide inside two weeks. Deletes it if the map near campus is empty.

WHAT BUSYBEE HAS TO DO

Live map with a capacity state, not just a wait in minutes, and a report that completes in one tap, because if it takes longer she is already walking.

NOTES ON THE WALL BEHIND THIS PERSONA

Restaurant yes, grocery store no.

● S7 · Q3

Popular times is a vibe: right about the shape, blind to today.

● S2 · Q4

I take my chances; if it's too busy I go somewhere else.

● S7 · Q5

It's an “aw man” or “dang it” kind of feeling.

● S7 · Q7

Almost every day: gym six days a week, store weekly, work, long drives.

● S7 · Q1

Downtown matters because of traffic and parking, not the venue itself.

● S7 · Q3



Sam The reciprocal reporter

Built from Subjects 3, 5 and 6 · Themes T3, T8, T9

SNAPSHOT

Already uses Waze and GasBuddy and accepts the bargain those apps make. Warns people he is meeting, gives a prediction when asked, and has never once written a review. He does not think of any of that as reporting.

GOALS

- Not be the reason someone else wastes a trip
- Contribute without attaching his real name to anything
- Use a tool that is already populated enough to be worth opening

FRUSTRATIONS

- Posting for strangers has no visible payoff, so who is even reading it?
- Won't trust a crowd whose moderation story is unexplained
- An app with nobody posting is just an empty map

WHAT THEY ALREADY DO

- Tells people before they head over, every time
- Reports traffic and conditions on Waze already, out of habit
- Assumes good faith, seeing no reason for anyone to call a busy place quiet

"I've definitely talked about how busy places are. If I knew someone else was thinking about going there, I might tell them how crowded it was."

Subject 6

WHERE THEY WALK AWAY

About 30 min at a restaurant, but will still enter a busy store to get what he needs

WHAT EARNS THEIR TRUST

Needs the moderation policy stated plainly, and needs to see the community is already active.

WHAT BUSYBEE HAS TO DO

Anonymous bee-name identity removes the exposure objection, and a reciprocity receipt, showing who your report actually helped, answers "what am I getting out of it."

NOTES ON THE WALL BEHIND THIS PERSONA

If someone asks about a restaurant, I'll give them my prediction.

● S5 · Q9 · recorded

Yes, when a friend is meeting up or going separately.

● S3 · Q9 · recorded

I see no reason for someone to rate a place not busy when it is.

● S3 · Q10 · recorded

If they're coming to meet up I'll tell them before they head over.

● S5 · Q9 · recorded

Probably, as long as it's run well and the information seems accurate.

● S6 · Q10

If I knew someone was thinking of going, I might tell them.

● S6 · Q9

WHAT THE INTERVIEWS CHANGED

This persona was revised by the Phase 2 data. The earlier draft cast Sam as a leaderboard-and-streak completionist. No participant raised gamification, and three questioned the payoff of contributing at all (Subjects 1, 2 and 4). The motive the interviews actually support is reciprocity, and the barrier is identity exposure, so badges and streaks are retention polish, not the reason anyone adopts.